

Jun Woo Beck

Audio & Music Machine Learning Researcher · Music Informatics Group, Georgia Tech

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RESEARCH SUMMARY

Master's student in Music Technology at Georgia Tech (Music Informatics Group, advisor Dr. Alexander Lerch) researching machine learning and music information retrieval for sound effects and music audio. First author of a DAFx 2026 paper on sound-effects dataset unification; planned thesis on multimodal music recommendation for video. Combines professional audio engineering experience and Berklee-trained musicianship with computational methods — bringing deep music and sound-design domain knowledge to audio ML research.

RESEARCH INTERESTS

Sound effects & music-audio machine learning · music information retrieval (MIR) · multimodal learning (audio, video, language) · audio dataset construction · interactive & biofeedback audio (secondary)

EDUCATION

Georgia Institute of Technology

Master of Science in Music Technology

Music Informatics Group · Advisor: Dr. Alexander Lerch

Relevant coursework: Machine Learning, Audio Content Analysis, Integrating Music into Multimedia

Atlanta, GA

Aug 2025 – Expected May 2027

Berklee College of Music

Bachelor of Music in Electronic Production and Design

Minors: Game and Interactive Media Scoring; Screen Scoring

Magna Cum Laude · GPA 3.71 / 4.00

Relevant coursework: Digital Signal Processing for Music Production and Postproduction, Music Acoustics

Boston, MA

Aug 2024

PUBLICATIONS

Peer-Reviewed Conference Proceedings

Beck, J. W., & Lerch, A. (2026). *Sound Effects Dataset Unification with the Universal Category System*. In Proc. International Conference on Digital Audio Effects (DAFx 2026), Boston, MA. [accepted; oral presentation, Sept 2026]

Oh, E. J., **Beck, J. W.**, & Smith, A. (2026). *The Singing Skin: An Audience-Centered Biofeedback System for Musical Interaction Based on Galvanic Skin Response*. In Proc. International Computer Music Conference (ICMC), Hamburg, Germany. [in press]

RESEARCH EXPERIENCE

Music Informatics Group, Georgia Institute of Technology

Graduate Researcher — Advisor: Dr. Alexander Lerch

Research spanning sound-effects dataset unification (DAFx 2026) and an industry-sponsored acoustic evaluation with Textron; thesis on multimodal music recommendation for video beginning Fall 2026.

Atlanta, GA

Aug 2025 – Present

SELECTED PROJECTS

Multimodal Music Recommendation for Video — Master's Thesis (planned)

From Fall 2026

- Thesis project to develop a multimodal model recommending fitting music for silent video by combining visual and text inputs to produce natural-language music recommendations.

Sound Effects Dataset Unification — EnvSound-UCS (DAFx 2026)

Fall 2025 – Spring 2026

- Built a rule-based tag-to-UCS conversion framework reaching 98.5–100% automatic conversion across FSD50K, AudioSet, and ESC-50.

- Released EnvSound-UCS, a unified 58,057-clip environmental-sound corpus spanning 59 categories, and validated it with audio-classification benchmarks.
- Open-sourced the conversion pipeline, tools, and converted datasets across four public repositories (github.com/JunWooBeck).

Textron Panel Project — Acoustic Evaluation & Digital EQ

Spring 2026

Industry collaboration with Textron · Music Informatics Group, Georgia Tech

- Conducted controlled acoustic measurements (THD, SINAD, frequency response) of speaker-panel audio quality in an isolation room against a consumer-laptop reference.
- Designed parametric EQ tuning presets that substantially reduced measured harmonic distortion and flattened the frequency response.

The Singing Skin — Audience-Centered Biofeedback System (ICMC 2026)

Fall 2025

- Designed and implemented the full audio system translating real-time galvanic skin response (GSR) signals into interactive musical output.

HONORS & AWARDS

GTCMT Seed Grant (\$2,000) — Georgia Tech Center for Music Technology

2026

Magna Cum Laude — Berklee College of Music

2024

Dean's List (4 semesters) — Berklee College of Music

2020 – 2023

SKILLS

Programming: Python, Max/MSP

Machine Learning: model training & evaluation; audio dataset construction & curation

Audio Engineering: recording & mixing (Pro Tools, Logic Pro, iZotope RX)

Sound Design: synthesis and sound-effects design for games, film, and video (FMOD, Wwise)

Music: composition, string arranging & orchestration; violin performance (classical & jazz)

MUSIC PRODUCTION & PERFORMANCE

Performer — Bio Sing-Along (GTCMT Seed Grant project)

2026

Performed at PRISM Pt. 1: Sound Through New Lens, Georgia Tech Center for Music Technology (May 2, 2026); designed the sound-morphing and real-time audio-processing system.

Recording / Mixing Engineer — Korean Ministry of Culture, Sports and Tourism

2021

Mixed and mastered orchestral music featuring traditional Korean instruments under music director Echae Kang (Seoul).

Recording / Mixing / Live Sound Engineer — Seoul Ballet Festival

2021

Recorded and mixed a full string-orchestra performance at the Seoul Arts Center; designed and mixed sound effects for live dance performances.

String Arranger — Dear Jazz Orchestra, "Sing for You Vol. 1"

2020

Arranged strings for the album's second track within a collaborative arrangement process.

Violinist — Dear Jazz Orchestra

2017 – 2022

Featured as soloist and orchestra member at the Seoul Jazz Festival and Jarasum Jazz Festival; recorded on multiple orchestra releases (Seoul).

Violinist — Berklee Orchestras

2020 – 2022

Berklee World Strings, Berklee Motion Picture Orchestra, and film-scoring session orchestras (Boston).